

Thesis for Doctor of Philosophy

Preparation and Mechanical
Properties of TPS(Thermoplastic
Starch)/Poly(Butylene Succinate)
Blends

June 2022

Graduate School
Kumoh National Institute of Technology

Department of Mechanical System Engineering

Hong Gil Dong

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(English summary)<-delete

Preparation and Mechanical Properties of TPS(Thermoplastic Starch)/Poly(Butylene Succinate) Blends

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Abstract

Polymer materials are widely used in industry nationwide.; however, most of these are non-biodegradable, and it is very difficult to manage their waste after use. These wastes become white pollution as a worldwide environmental problem. Development of a biodegradable polymer is one of the solutions to abate the aforementioned problem. Biodegradable polymers can be developed by combining biodegradable matrices and biodegradable fillers.

(Korean summary))<-delete

TPS(Thermoplastic Starch)와 Poly(Butylene Succinate)의 Blends 제조와 물리적 특성

홍길동

국립금오공과대학교 대학원 기계시스템공학과

요약

본 연구에서는 대표적인 천연 고분자의 하나인 전분을 여러 가소제를 사용하여 가소화 시킨 thermoplastis starch(TPS)를 제조하여 이를 지방족 폴리에스터의 하나인 polybutylene succinte(PBS)와 혼합비를 달리하며 블렌드 하였다. TPS의 조성 및 함량이 PBS의 열적 특성 및 물리적 특성에 미치는 영향에 대해 확인하였다.

본 연구에서는 대표적인 천연 고분자의 하나인 전분을 여러 가소제를 사용하여 가소화 시킨 thermoplastis starch(TPS)를 제조하여 이를 지방족 폴리에스터의 하나인 polybutylene succinte(PBS)와 혼합비를 달리하며 블렌드 하였다. TPS의 조성 및 함량이 PBS의 열적 특성 및 물리적 특성에 미치는 영향에 대해 확인하였다.

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